LESSON PLAN

AIRCRAFT PROPULSION & ENGINEERING SYSTEM (6)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	03
Total No. of Lesson Plans	14	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Various types of corrosion and their identification
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Various types of corrosion and their identification.
5. Specific Objectives	6.1.1.1 Trainee will be able to describe types of corrosion.

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Identification of corrosion types using sample coupons; Safety: Wear gloves and eye protection; Supervision: Instructor monitors sample handling and disposal.
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What is the primary cause of corrosion in aircraft structures?
(b) Name one environmental factor that accelerates corrosion.
(c) Why is corrosion considered a silent enemy in aviation maintenance?

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Galvanic corrosion mechanism • Crevice corrosion formation • Pitting corrosion characteristics • Intergranular corrosion causes • Fretting corrosion definition	L & Ds	PPT, IB	15 min
(b)	• Stress corrosion cracking signs • Exfoliation corrosion features • Uniform corrosion patterns • Sample coupon examination • Visual identification criteria	L & Ds	PPT, PDF	20 min
(c)	• Differentiating corrosion types • Documentation of findings • Reporting procedures • Preventive measures overview • Common misconceptions review	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain the difference between galvanic and crevice corrosion.
(b) Describe the visual indicators of pitting corrosion.
(c) How does stress corrosion cracking differ from uniform corrosion?

12. Home Assignment and Follow Up	Review LP 02 on protective coatings and list three specific measures that prevent the types of corrosion discussed in this lesson.

Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Short term protective and precautions while using short term protective.	Refer to the master syllabus reference list	LP 04

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min